

Parabolas

Montague Hamilton

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1 Questions

1. **Describe fully and test:** $y = -3(x + 2)^2 + 5$ - **vertex, opening direction, shape and the y-intercept.**

The vertex will occur at $(-2, 5)$, as

$$y = -3((-2) + 2)^2 + 5$$

$$y = -3(0)^2 + 5$$

$$y = 5$$

The opening direction will be downwards, as the y values get inverted by the negative in front of the expression, $-3(x + 2)^2$. The shape will be quite narrow, as for every addition to x, y changes by 3 (shrinks in this case). The y-intercept occurs when $x = 0$, so

$$y = -3((0) + 2)^2 + 5$$

$$y = -3 \cdot 4 + 5$$

$$y = -7$$

The y-intercept occurs at $(0, -7)$.

2. **Describe fully:** $y = 2 \sin(3x) - 4$ - **amplitude, midline and period.**

The amplitude will be 2, as for every value of $\sin(3x)$, y is twice as big as otherwise. The midline will be -4 , as the last number added or subtracted from the rest of the function shifts every y output by that amount. The period will be 6π , as the function will cycle 3 times faster than otherwise.

3. **Spot the error: a student says $y = (x + 3)^2$ has vertex $(3, 0)$. Test the claim, state the true vertex, and name the rule they violated.**

At $x = 3$,

$$y = ((3) + 3)^2$$

$$y = 36$$

At $x = 2$, $y = 25$ and at $x = 4$, $y = 49$, so it is clear that $x = 3$ can't be the vertex, as the y values for $x = 2$ and $x = 4$ aren't the same. The vertex will be at $x = -3$, as

$$y = ((-3) + 3)^2$$

$$y = (0)^2$$

To check, $x = -4$, $y = 1$, should be equal to, $x = -2$, $y = 1$. So the vertex is at $(-3, 0)$. The student most likely forgot the negative in front of the 3.